

Hasta-Shilpa

Design Bridge for Bamboo & Cane Artisans

Self-Employment · Western Ghats · Sustainable Craft

Platform	Domain	Category	Material	AI Feature
Android	Artisan Tech	Self-Employment	Bamboo / Cane	Price Suggester

1. Problem Statement

Bamboo and cane artisans in the **Western Ghats** possess centuries of craft knowledge and produce high-quality furniture, baskets, and home goods. Despite their skill, they face a critical challenge: **design stagnation**. Artisans continue producing the same traditional patterns that no longer align with modern urban consumer preferences, leading to low sales volumes, reduced incomes, and the gradual decline of a rich heritage craft.

Requirement	Description
Design Gap	No access to modern design trends or product inspiration libraries.
Measurement Gap	Lack of standardised blueprints leads to inconsistent product quality.
Pricing Gap	Artisans cannot accurately price products; often undersell.
Market Gap	No direct channel to urban buyers; dependent on exploitative middlemen.
Material Tracking	No system to track raw material usage per batch or calculate unit cost.

2. Scope of the App

In Scope

- Curated design inspiration library for bamboo and cane products.
- Downloadable blueprints (PDF/image) with measurements and material notes.
- Material usage tracker per production batch.
- Intelligent price suggester (material cost + labour hours → suggested price).
- Simulated marketplace for listing and viewing finished products.
- Trending product feed showcasing market-relevant bamboo/cane items.

Out of Scope

- ✗ Live payment processing or e-commerce transactions.

- ✗ Real-time logistics or shipping integration.
- ✗ Multi-vendor platform management (admin panel).
- ✗ Inventory management across supply chain.

3. Functional Requirements

Requirement	Description
FR-01 Design Library	Display a scrollable, searchable feed of modern bamboo/cane product photos organised by category.
FR-02 Blueprint View	Show detailed sketch/diagram with labelled measurements (cm/inches). Must be pinch-zoomable.
FR-03 Download Blueprint	Export blueprint as PDF or image to device storage for offline reference.
FR-04 Trending Feed	Show trending products (e.g., Bamboo Laptop Stand, Modern Lamp Shade) updated from a curated list.
FR-05 Material Tracker	Allow artisan to log raw material (bamboo poles, cane bundles) used per batch and store history.
FR-06 Price Suggester	Accept inputs: material cost (₹), labour hours, overhead %; output: minimum and suggested selling price.
FR-07 Simulated Listing	Artisan can create a product listing with photo, name, description, and suggested price for preview.
FR-08 Category Filter	Filter design library by product type: furniture, storage, lighting, décor, utility.
FR-09 Search	Full-text search across design library by product name or keyword.
FR-10 Onboarding	First-launch tutorial explaining all four main features in visual step-by-step screens.

4. Non-Functional Requirements

Requirement	Description
NFR-01 Performance	App must load the design feed within 2 seconds on a 4G connection.
NFR-02 Offline Mode	Core design library and saved blueprints accessible without internet.
NFR-03 Usability	UI must be navigable by users with low digital literacy; icon-driven, minimal text inputs.
NFR-04 Scalability	Architecture must support addition of 500+ new design entries without code changes.
NFR-05 Accessibility	Minimum text size 14sp; high-contrast mode; support for TalkBack screen reader.
NFR-06 Storage	App should not exceed 50 MB installed size; images cached efficiently.
NFR-07 Security	No sensitive personal data stored; local data encrypted with Android Keystore.
NFR-08 Compatibility	Must support Android 8.0 (API 26) and above, covering ~95% of target devices.
NFR-09 Localization	String resources externalised to support future Kannada / Telugu translation.
NFR-10 Maintainability	Codebase follows MVVM architecture; unit test coverage ≥ 60% for business logic.

5. App Usage & User Flow

#	Step	Action
Step 1	Launch & Onboard	Artisan opens app → views 3-screen visual tutorial → reaches Home Dashboard.
Step 2	Browse Trends	Trending feed displays modern bamboo/cane products. Artisan taps a card to view details.
Step 3	View Blueprint	Detail screen shows product photo + zoomable blueprint with measurements (e.g., 30 cm × 15 cm).
Step 4	Download Blueprint	Artisan taps Download → selects PDF or Image → saved to device for workshop use.
Step 5	Log Materials	Opens Material Tracker → enters batch details (e.g., 12 bamboo poles) → saves entry.
Step 6	Calculate Price	Opens Price Suggester → enters material cost (■) + hours worked → app outputs price range.
Step 7	Create Listing	Artisan uploads product photo, adds name and price → listing appears in simulated marketplace.

6. Features

■ Must-Have Features

- ✓ Design Inspiration Library (card-based, categorised)
- ✓ Zoomable Blueprint View with measurements
- ✓ PDF / Image Blueprint Download
- ✓ Material Usage Tracker (per batch logging)
- ✓ Price Suggester (material cost + labour → price)
- ✓ Trending Products Feed
- ✓ Simulated Product Marketplace Listing
- ✓ Category & Search Filters
- ✓ Visual Onboarding Tutorial

■ Good-to-Have Features

- ◆ Offline blueprint access & caching
- ◆ Artisan community gallery (share designs)
- ◆ Regional language UI (Kannada, Telugu)
- ◆ AI-powered seasonal design suggestions
- ◆ QR code on listing for easy sharing
- ◆ Video tutorial clips alongside blueprints
- ◆ Batch history & analytics dashboard
- ◆ Barcode scan for material entry

7. Technical Implementation

Requirement	Description
Language	Kotlin (primary) with Java interop where required.
Architecture	MVVM (Model-View-ViewModel) with LiveData and ViewModel components.
UI Framework	Jetpack Compose or XML layouts with RecyclerView + CardView for design feed.
Image Loading	Glide or Coil library for efficient image loading and caching.
Local Storage	Room Database for material tracker logs and blueprint metadata.
File Export	Android PdfDocument API for blueprint PDF generation; MediaStore for image save.
Blueprint Zoom	PhotoView or custom GestureDetector for pinch-to-zoom on blueprint images.
Price Logic	ViewModel-layer pure Kotlin calculation: (material cost + labour cost) × margin %.

Requirement	Description
Design Assets	Hosted on Firebase Storage or bundled in assets/ folder (< 50 MB APK target).
Navigation	Jetpack Navigation Component with bottom navigation bar (4 tabs).
Testing	JUnit 4 for unit tests; Espresso for UI tests; Mockito for ViewModel mocking.

8. Impact Goals

Impact Area	Description
■ Artisanal Modernization	Keeps traditional bamboo/cane crafts relevant in 21st-century urban markets by bridging design gaps.
■ Economic Growth	Increases the per-unit value of an artisan's work through modern, market-aligned product designs.
■ Sustainable Material	Promotes bamboo as a green alternative to plastic and metal, supporting eco-conscious consumption.
■ Digital Inclusion	Brings rural artisans into the digital economy with a simple, visual-first mobile interface.
■ Market Access	Simulated marketplace gives artisans exposure to modern pricing and direct-to-consumer channels.

9. Success Criteria for Students

Criterion	Priority	Description
Blueprint Clarity	Must	Blueprints are zoomable, high-resolution, and printable.
Price Suggester	Must	Allows input of local material cost + hours; outputs suggested selling price.
Visual UI	Must	Clean card-based layout focused on photos and visual learning.
Download Blueprint	Must	PDF/Image export from within the app for offline use.
Material Tracker	Must	Log bamboo poles/cane bundles per production batch accurately.
Marketplace Listing	Should	Simulated product listing with photo, name, and price.
Offline Access	Should	Core design library accessible without active internet.
Multi-language UI	Could	Kannada / regional language support for artisans.
AI Design Suggest	Could	AI-curated trend suggestions based on season/market.

10. The Vision — Hasta-Shilpa as a Design Bridge

Hasta-Shilpa (meaning "Hand Craft" in Sanskrit) is conceived as more than an app — it is a **Design Bridge** that connects the centuries-old wisdom of Western Ghats artisans with the aesthetic sensibilities of 21st-century urban consumers. By placing a curated design library, blueprint tools, and a price calculator directly in an

artisan's hands via a simple Android app, the project aims to restore economic dignity to skilled craftspeople while preserving and modernising India's bamboo and cane heritage.

Students building this project will gain hands-on experience in **end-to-end Android development**, including file I/O, database management, business logic implementation, and UI/UX design for low-literacy users — all within the meaningful context of social impact technology.

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